

AI-Powered MERN Stack E-Commerce Platform

Overview

This project is a **modern AI-powered full-stack e-commerce platform** built using the **MERN stack (MongoDB, Express.js, React, Node.js)** with advanced features like **semantic product search, multi-role dashboards, and secure payment integration**.

The platform simulates a **real production-level marketplace** where buyers can browse and purchase products while sellers manage their inventory through a dedicated dashboard.

The system also integrates **AI capabilities** to enable intelligent product search and recommendations.

★ Key Features

☐ AI Semantic Product Search

- AI powered search that understands **user intent instead of just keywords**
- Uses **vector embeddings for product recommendations**

Multi-Role Authentication System

- Secure **JWT authentication**
- Three user roles:
 - **Buyer**
 - **Seller (Vendor)**
 - **Admin**

Seller Dashboard

- Add, edit, and manage products
- Manage stock and categories
- View seller analytics
- Upload product images

Buyer Features

- Browse products by category
- Wishlist system
- Add to cart
- Checkout with secure payment

Stripe Payment Integration

- Secure checkout system
- Test payments supported

Smart Product Discovery

- Category filtering
- Product search
- AI product recommendations

Admin Control Panel

- Manage categories
- Manage platform data
- Secure admin access

Tech Stack

Frontend

- React
- TypeScript
- Vite
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas
- Mongoose ODM

AI Integration

- Google Gemini API
- Embedding-based semantic search

Payments

- Stripe API

Architecture

The application follows a **modern scalable architecture**:

Frontend (React)



REST API (Express.js)



Database (MongoDB)



AI Services (Gemini API)

This structure ensures **scalability, maintainability, and performance**.

Core Modules

- Authentication System
- Product Management
- Category Management
- Wishlist System
- Shopping Cart
- Order Processing
- AI Recommendation Engine
- Vendor Dashboard
- Admin Panel

Security

- JWT authentication

- Protected routes
 - Role-based access control
 - Secure API structure
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Project Goal

The goal of this project was to build a **real-world scalable e-commerce system** while integrating **AI-driven product discovery** and modern development practices.

Developed By

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Specializing in:

- Full-Stack Web Development
- AI-Powered Applications
- Scalable MERN Architecture